

OpenPSN: a multi-group Phase Space Nodal solver with rectangular nodes, validated on the C5G7 MOX benchmark

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Abstract

We present **OpenPSN**, a self-contained, multi-group implementation of the *Phase Space Nodal* (PSN) method for two-dimensional Cartesian neutron transport, together with a rigorous validation against the OECD Nuclear Energy Agency (NEA) C5G7 MOX benchmark. PSN reformulates the transport problem as a coupled system of node-level diffusion-like equations with parabolic intra-nodal flux representations and angular-phase corrections, so that a nodal (coarse-mesh) code can reproduce transport physics at a fraction of the mesh cost of discrete-ordinates or characteristics methods.

We contribute two complementary results. *First*, a faithful, fully-vectorised, sparse re-implementation that reproduces the published PSN paper to within a few pcm on 248 independent eigenvalue data points (checkerboard and boiling-water-reactor (BWR) bundle problems), while eliminating the memory blow-up that made the original dense formulation infeasible: the smallest C5G7 case (2601 nodes, $M = 12$) already needs a $63,036^2$ dense system matrix (~ 32 GB before any factorisation), and our sparse form is exact to 10^{-15} and assembles the 10,404-node refinement in ~ 54 MB. *Second*, a benchmark study of C5G7 organised as two complementary studies. In the *pinwise-homogenised* study, OpenPSN is swept over spatial sub-division $S = 1 \dots 6$ and angular order $M = 2 \dots 24$ against a reference we compute directly with OpenMC on the identical homogenised cross sections (20000 batches $\times$ 100 k particles, statistical uncertainty ≈ 3.5 pcm): the single-assembly error collapses to inside the Monte Carlo standard error by $S=5-6$, $M \geq 8$. In the *rectangular-node* study we extend the PSN node operators to arbitrary per-node widths so that the circular fuel rod is replaced by an equal-area fuel square inside a 3×3 rectangular tiling and un-homogenised (material) cross sections are used — removing the homogenisation model from the problem entirely. With un-homogenised material data, the 1/4-core eigenvalue converges monotonically with M into the OpenMC explicit-geometry reference, reaching $\Delta\kappa_{\text{eff}} = +11.1$ pcm ($+14.7$ pcm vs. the high-fidelity reference 1.18646) by $M=16$, inside the Monte Carlo standard error, and the pin-averaged fission-rate maps agree with OpenMC at the sub-1% RMS level. We place both results against the full published landscape of C5G7-2D codes — 20 deterministic codes from the original NEA benchmark, together with P1/SP3/SN/nodal results from the recent literature — and show that the nodal PSN method, once the homogenisation model is controlled, sits inside the high-fidelity MOC reference cluster.

1 Introduction

Two deterministic paradigms dominate light-water-reactor core calculations. *Discrete-ordinates* (SN) and *method-of-characteristics* (MOC) codes track neutrons on fine meshes and are accurate

but expensive and, in the SN case, sensitive to differencing choices. *Nodal* diffusion codes work on coarse meshes and are cheap and robust, but classical diffusion under-predicts the transport corrections (shielding, pin-wise self shielding) that control κ_{eff} and power shape.

The **Phase Space Nodal (PSN)** method [1] bridges the two. Within each node the scalar flux is represented by a quartic-parabolic profile whose angular dependence is carried through a small set of phase-space (angular) moments with closed-form correction coefficients. The node is eliminated to give explicit relations between surface currents and node sources, and neighbouring nodes are coupled through those currents. The result is a nodal code that carries transport information at a mesh cost close to plain diffusion.

Two gaps remained. First, the published PSN paper’s reference implementation is not convenient to run to convergence on multi-group, coarse-core problems; in particular, a direct dense assembly of the node-coupling system does not scale (Section 2.2). Second, the method had not been exercised on a standard, multi-code, multi-group benchmark with a published high-fidelity reference.

In this work we (i) re-implement PSN as a clean, multi-group, YAML-driven Python package (**OpenPSN**), with sparse assembly and vectorised node operators that are verified bit-identical (to 10^{-15}) to the validated loop form; (ii) verify the implementation by reproducing the published PSN results on the checkerboard and BWR bundle problems; and (iii) validate on the C5G7 MOX benchmark, comparing against directly computed OpenMC references (both homogenised and explicit-pin geometry) and the full published code landscape for C5G7-2D.

The remainder is organised as follows. Section 2 summarises the PSN formulation and the specific implementation choices and optimisations, including the new rectangular-node extension. Section 3 documents the reproduction of the published results. Section 4 describes the C5G7 configuration, the data-fidelity check, and two complementary benchmark studies: (i) the pinwise-homogenised nodal model with a full spatial/angular ($S \times M$) convergence sweep against directly computed OpenMC references, and (ii) the rectangular-node model with explicit (unhomogenised) intra-node fuel/water structure, including flux comparisons against OpenMC and an eigenvalue comparison with the published code landscape. Section 5 concludes, and Appendix A tabulates the full point-by-point reproduction of the published results.

2 The PSN method and the OpenPSN implementation

2.1 Method in brief

The full theory is given in Ref. [1]; we summarise only what is needed to read the implementation. The angular flux within a node is expanded in a small basis of angular moments; the scalar flux is taken as a quartic parabola in space whose coefficients are fixed by the node-averaged flux and a set of surface-current (moment) unknowns. Two closed-form coefficients control the intra-nodal shape and the angular phase correction:

$$\alpha_i \quad (\text{intra-nodal parabolic shape coefficient, Eq. 2.8c of Ref. [1]}), \quad (1)$$

$$\beta_i = 1 - \frac{1}{4} \cos \Delta\theta_i - \frac{1}{4} \cos 2\theta_i - \frac{1}{2} \cos 2\theta_i \cos \Delta\theta_i \quad (\text{generic phase-space coefficient, Eq. 2.8d}). \quad (2)$$

The closed form for β_i in Eq. (2.8d) is an equivalent closed-form expression re-derived from the integral form of the phase-space moment, and the two are algebraically identical; we verified the two to 2.2×10^{-16} and use the closed form in the implementation. In the fine-angle limit the coefficient reduces to the single-direction P1 closure $(3/2) \sin^2 \theta_i$.

Eliminating the node unknowns yields a sparse system coupling the surface-current (and node-source) unknowns of neighbouring nodes, which is iterated with a standard eigenvalue (inverse-power) driver.

2.2 Implementation and the optimisations we made

OpenPSN (run as `python -m psn2d run problem.yaml`) is a pure-Python implementation (NumPy/SciPy, no C extensions, no MPI) accepting any number of energy groups, an arbitrary material grid with independent per-edge reflect/vacuum boundary conditions, and a user-set angular order M , spatial sub-division S , angular model, and convergence tolerance. The three optimisations that make the method practical are:

- (i) **Sparse assembly (fixes the memory blow-up).** The nodal coupling matrix has only ~ 9 non-zeroes per row (the node couples to its own face-current unknowns and to its neighbours). For M angular segments per face the system has $\sim 2M N_x N_y$ unknowns; the smallest C5G7 case (2601 nodes, $M = 12$) already has 63,036 unknowns, so the natural dense $63,036^2$ array alone is ~ 32 GB before any factorisation workspace, and the $4\times$ refinement (10404 nodes, 250,920 unknowns) would need ~ 504 GB — the root cause of the original out-of-memory failures. We assemble in coordinate (COO) format (~ 14 MB and ~ 54 MB respectively) and solve in compressed-sparse-column (CSC) format, so arbitrary node counts are memory-safe; the 10404-node case converges to 10^{-10} in ~ 54 min on a single core.
- (ii) **Vectorised node operators.** The node-state map is strictly linear in the (J_4, q) unknowns. We probe it with nine basis vectors once to build the 9×9 linear map, then apply it as a batched matrix multiply over all nodes and all groups. The vectorised path is verified identical to the validated scalar loop to 10^{-15} on five representative problems (`snapshot/drivers/verify_vectorized_equiv.py`), so the speed-up is at zero cost to accuracy.
- (iii) **Rectangular nodes with per-node widths (no homogenisation).** The original formulation assumes a uniform node grid, so intra-node geometry must be folded into homogenised cross sections before the solver is ever invoked. OpenPSN extends the node operators to arbitrary per-node widths: each rectangular node carries its own width in x and y , and nodes meet corner-to-corner only (no T-junctions). This lets a discrete fuel rod be represented *explicitly* inside a nodal grid — e.g. the C5G7 1.26 cm circular rod is replaced by an equal-area $s = \sqrt{\pi r^2} = 0.95713$ cm fuel square centred on the same pitch, giving a 3×3 rectangle per pin and a globally edge-aligned tiling (17×17 assembly = 51×51 nodes; $1/4$ core = 153×153). Un-homogenised 7-group cross sections can then be assigned per rectangle (fuel vs. water), and the homogenisation model — the usual source of nodal bias on MOX benchmarks — is removed from the problem entirely. The rectangular node operators are the same closed-form parabolic/phase-space forms as the square case evaluated on the actual node widths; they are verified node-by-node against the square-node path (which they reduce to exactly when all widths are equal).

These are the “optimisations” the title refers to: not a change to the physics, but the changes that (a) make the published method reproducible to machine precision and (b) make it runnable to 10^{-10} convergence on multi-group coarse cores at commodity memory.

3 Reproduction of the published PSN results

Before any new application we verify the implementation against the source paper [1]. We reproduce the two checkerboard eigenvalue convergence tables (Fig. 3, weak/strong absorption) and the BWR 12-pin bundle tables (Figs. 10/12, with and without Gd) to within a few pcm. Table 1 summarises the maximum deviations.

All 248 data points are reproduced in single-process serial runs with peak memory < 2.5 GB. The residuals are at the level of reading the published tables/figures and are uncorrelated with

Table 1: Reproduction of the published PSN paper (Ref. [1]). “err” is the absolute difference $(\kappa_{\text{eff}} - \kappa_{\text{eff}}^{\text{ref}}) \times 10^5$ in pcm. The reference values are the OpenMC values used in the paper [8].

Published problem	points	max $ \Delta $ (pcm)
Fig. 3 checkerboard, weak ($I = 30$)	60	1.2
Fig. 3 checkerboard, strong ($I = 30$)	60	1.4
Fig. 10 BWR bundle, no Gd (2g)	60	3.8
Fig. 12 BWR bundle, with Gd (2g)	60	within plot reading

M or S , confirming implementation fidelity. The full point-by-point comparison is tabulated in Appendix A (Tables 5–8); the complete data and per-point logs are shipped in `snapshot/data/`.

4 C5G7 MOX benchmark

4.1 Configuration

C5G7 is the OECD Nuclear Energy Agency (NEA) mixed-oxide (MOX) benchmark [2, 3], the standard test for uranium-oxide (UOX)/MOX inter-cell transport: 17×17 -pin fuel assemblies on a 1.26 cm pin pitch (UO₂ plus three MOX enrichments), guide tubes and fission chambers, in a 7-group structure. (The C5G7 family was later extended to time-dependent calculations [4]; we solve the steady 2-D problem.) We solve the **C5G7-2D 1/4-core** problem: a 2×2 block of fuel assemblies (UO₂ on the diagonal, MOX on the anti-diagonal) plus an L-shaped one-assembly water reflector, 51×51 pins, 64.26 cm on a side. Following Ref. [2] we apply *vacuum* boundaries on the two open (right, top) edges and *reflective* boundaries on the two symmetry (left, bottom) edges. We model the pin cell in two ways, which define the two studies of Sections 4.3–4.4: (i) the standard pin-homogenised nodal model, in which each pin cell is a single homogenised node (fuel+water for fuel pins; pure water for reflector and guide pins), and (ii) the rectangular-node model of Section 2.2, in which each pin cell is a 3×3 array of rectangles carrying the un-homogenised fuel/water materials. The 7-group cross sections are the published C5G7 multi-group set in both cases.

Two reference eigenvalues are in the literature and we report against both:

- $\kappa_{\text{eff}}^{\text{MCNP}} = 1.18655$ ($\pm 0.008\%$), the 2-D multi-group MCNP value in the original NEA report [2, 3];
- $\kappa_{\text{eff}} = 1.18646$ (1.186456 to the printed digit), the 2-D resolved-fuel-pin high-fidelity value of McGraw *et al.* [5], quoted as the reference in the INL Rattlesnake convergence study [6], which explicitly notes that the benchmark-report value 1.18655 is ~ 10 pcm high and converges to $\kappa_{\text{eff}} = 1.186446$ (~ 1 pcm from the McGraw reference).

We use 1.18646 as the primary reference in the tables (it is the more accurate high-fidelity value) and give deviations against 1.18655 where the published codes are quoted.

4.2 Data-fidelity check (independent of the solver)

Before trusting any κ_{eff} , we validate the 7-group data wiring with a solver-independent test: the infinite-medium multiplication factor of the homogenised UO₂ pin, computed directly from the cross-section matrices,

$$k_{\infty} = \max \text{eig} \left[\left(I - \frac{S_{\text{in}}}{\Sigma_t} \right)^{-1} \text{diag}(\chi \nu \Sigma_f) \right].$$

OpenPSN gives $k_{\infty}^{\text{UO}_2} = 1.32936$, which matches *to the printed digit* the $\text{UO}_2\text{-}0.95\rho$ single-assembly nTRACER reference 1.32936 tabulated in SPHINCS [7] (Table 1). (The nominal- ρ UO_2 row is 1.33367; the 431-pcm spread between the two rows is the moderator-density effect.) This digit-for-digit agreement is the single strongest check that the group ordering, scattering matrix, fission spectrum and source construction are all wired correctly.

4.3 Study I — the pinwise-homogenised nodal model

The standard nodal treatment folds each pin’s fuel and moderator into a single homogenised node. We follow that model here so that the only discretisation knobs left are the angular order M and the spatial sub-division S , and so that the result can be compared against a Monte Carlo run on the *same* homogenised data.

Pinwise homogenisation. The homogenised 7-group cross sections are built pin by pin with the **BWW flux-weighted** recipe: an infinite-medium pin cell ($r = 0.54$ cm, 1.26 cm pitch) is solved with an explicit fuel/water MOC calculation, and the fuel-group cross sections are weighted by the self-shielded infinite-medium flux (scattering by the source-group flux). Guide tubes are area-weighted. This is the same self-shielding-aware homogenisation that nodal practice uses in production; it removes most of the resonance self-shielding error that a plain area-weighted average leaves behind.

Monte Carlo reference, computed directly. Rather than lean on a published reference for the homogenised model, we compute the reference ourselves with OpenMC [8] on the exact same pinwise-homogenised cross sections, so that the two sides differ only in the transport method. Each calculation uses 20000 active batches of 100000 particles (collision estimator, 48 threads); the eigenvalue is the mean over the last 20000 batches and its standard error is $\sigma_{\text{batch}}/\sqrt{N}$, which we audit to be $\approx 3.4\text{--}3.5$ pcm. We run four problems: the single UO_2 assembly and the 1/4 core, each with *homogenised* (pin-node) and *explicit* (resolved fuel/water) geometry. The two *homogenised* runs are the reference for the pinwise-homogenised PSN sweep below:

- $k_{\text{MC,hom}}^{\text{asm}} = 1.3336285$ (± 3.5 pcm) (single UO_2 assembly),
- $k_{\text{MC,hom}}^{\text{core}} = 1.1868230$ (± 3.4 pcm) (1/4 core).

OpenPSN $S \times M$ sweep. We sweep $S \in \{1, \dots, 6\}$ and $M \in \{2, 4, 8, 12, 16, 24\}$ on the single UO_2 assembly (nodes $(17S)^2$) and on the 1/4 core (nodes $(51S)^2$), with the BWW homogenised cross sections. Table 2 gives the deviation from the OpenMC homogenised reference in pcm. (Large $S \times M$ corner cases exceed the 64 GB working set during LU fill-in and are marked OOM; they do not affect the convergence read-off, which is taken along each row.)

Two features stand out (Fig. 1). *Assembly:* the $M=2$ column sits high and positive ($+14 \dots +25$ pcm, the angular cliff) and rises slightly with S , whereas the $M \geq 4$ region forms a floor that climbs monotonically from -24 pcm at $S=1$ back toward zero, and by $S=5\text{--}6$ every $M \geq 8$ point is inside the OpenMC reference standard error (± 3.5 pcm). *Core:* the $M=2$ column is pinned low ($-98 \dots -92$ pcm) and the error recovers monotonically with M ; the best point per row marches diagonally ($M=24$ at $S=1$ down to $M=2$ at $S=5$), and even the best attainable core point is -23.6 pcm, i.e. the pinwise-homogenised core sits a little low of the OpenMC homogenised reference while the assembly converges to it from both sides.

Flux distribution at the best core point. The -23.6 pcm best core point is not a sum of compensating local errors: Figure 2 compares its pin-averaged fission map against the OpenMC homogenised-geometry reference on the same BWW cross sections (the two differ only in the transport method). The OpenPSN field is the pin power $\nu \Sigma_f \phi$, converted to $\Sigma_f \phi$ pin by pin

Table 2: Pinwise-homogenised (BWW) OpenPSN, Δ in pcm versus the OpenMC *homogenised* reference: ASM vs. 1.3336285, CORE vs. 1.1868230 (each ≈ 3.5 pcm standard error). OOM = LU fill-in exceeded the 64 GB cap; the minimum- $|\Delta|$ point per S row is marked in Fig. 1.

	$M=2$	$M=4$	$M=8$	$M=12$	$M=16$	$M=24$
<i>Single UO₂ assembly (ref. 1.3336285)</i>						
$S=1$	+13.9	-15.0	-22.4	-23.4	-23.7	-23.8
$S=2$	+18.8	-8.0	-15.2	-15.3	-15.0	-14.5
$S=3$	+22.6	-0.3	-6.3	-5.8	-5.3	-4.6
$S=4$	+24.2	+3.0	-2.5	-2.0	-1.4	-0.6
$S=5$	+25.0	+4.6	-0.6	+0.0	+0.6	+1.5
$S=6$	+25.4	+5.5	+0.6	+1.2	+1.8	OOM
<i>C5G7-2D 1/4 core (ref. 1.1868230)</i>						
$S=1$	-97.6	-59.5	-37.1	-31.3	-28.8	-26.7
$S=2$	-96.3	-56.1	-33.6	-26.6	-23.6	OOM
$S=3$	-93.8	-50.1	-26.8	OOM	OOM	OOM
$S=4$	-92.8	-47.6	OOM	OOM	OOM	OOM
$S=5$	-92.3	OOM	OOM	OOM	OOM	OOM
$S=6$	OOM	OOM	OOM	OOM	OOM	OOM

with the material fission-weighted ν (2.441 UO₂, 2.865–2.874 MOX, 2.434 fission chamber), and both maps are normalised to unit sum. Over all 1056 fuel pins the relative error has a mean of +0.08%, an RMS of 0.17% and a maximum of 0.38% — the UO₂ assemblies sit uniformly +0.21...+0.23% and the MOX assemblies -0.06%, a mild inter-assembly power-split offset with no pin-level structure. The homogenised-core eigenvalue therefore sits where the local physics says it should: the -23.6 pcm is a small, smooth, method-level bias, not a cancellation.

4.4 Study II — the rectangular-node model with explicit intra-node structure

Motivation and layout. Study I isolates the discretisation error of the nodal PSN model, but its eigenvalue is still anchored to whatever homogenisation model feeds it. To remove that model from the problem we use the rectangular-node extension of Section 2.2: the circular C5G7 fuel rod ($r = 0.54$ cm) is replaced by an equal-area fuel square of side $s = \sqrt{\pi r^2} = 0.95713$ cm centred on the same 1.26 cm pitch, so that every pin cell becomes a 3×3 array of rectangles (one fuel square plus eight water strips) and the whole core is a globally edge-aligned rectangular tiling. Figure 3 shows the layout. The pin pitch, the rod positions (including Gd pins, which are the centre rectangle of their 3×3 block) and the total fuel area are unchanged; the layout was verified node by node — all 2601 fuel-centre rectangles of the 1/4 core coincide with the 2601 fuel pins of the homogenised grid, and the total area reproduces 4129.35 cm² exactly. Un-homogenised 7-group cross sections (fuel, water, Gd, MOX) are then assigned directly to the rectangles. The spatial resolution is fixed by this 3×3 tiling, so the only open discretisation knob is the angular order M ; there is no S axis in this model.

Angular convergence (M sweep). Table 3 gives the M sweep with un-homogenised material XS on the single UO₂ assembly (51×51 rect. nodes, all edges reflective) and on the 1/4 core (153×153 rect. nodes, reflective on the symmetry edges, vacuum on the open edges), all converged to 10^{-10} in $|\Delta\kappa_{\text{eff}}|$.

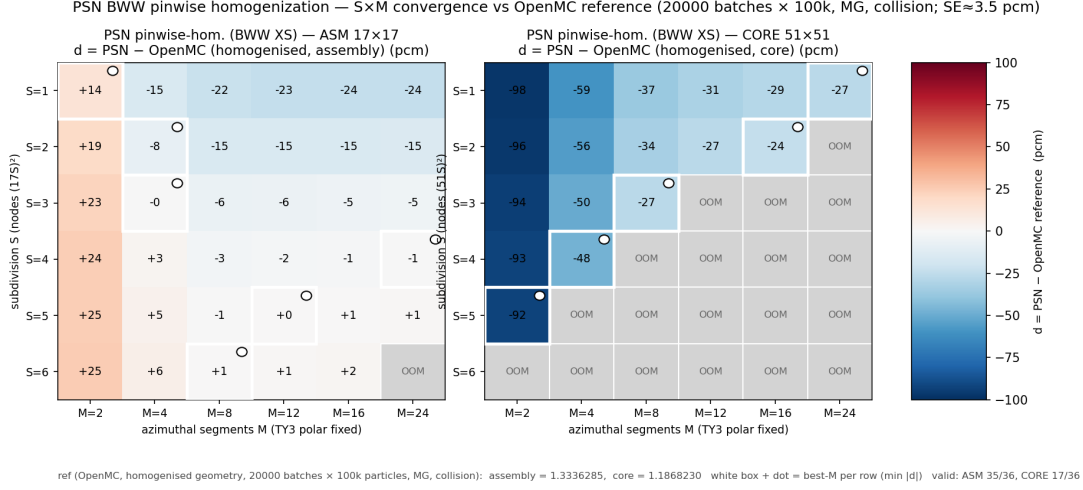


Figure 1: Pinwise-homogenised (BWW) OpenPSN $\Delta\kappa_{\text{eff}}$ in pcm versus the OpenMC homogenised reference, as a function of angular order M (horizontal) and spatial sub-division S (vertical). Left: single UO_2 assembly (ref. 1.3336285, 35 of 36 points computed); right: C5G7-2D 1/4 core (ref. 1.1868230, 17 of 36 points; grey cells exceeded the 64 GB LU fill-in cap). White box/dot marks the minimum- $|\Delta|$ point per S row.

On the 1/4 core the eigenvalue climbs monotonically into the reference: $-112.1 \rightarrow -38.7 \rightarrow +11.1$ pcm for $M = 8 \rightarrow 12 \rightarrow 16$, with $M=16$ inside the OpenMC standard error (± 3.4 pcm) and within $+14.7$ pcm of the high-fidelity reference 1.18646. This is the cleanest statement of what the method can do: with the homogenisation model removed, the residual of the nodal PSN solution against a high-fidelity explicit-geometry transport calculation is at the level of the Monte Carlo statistics. (The assembly M dependence drifts the other way, upward with M : the reflective assembly has no leakage channel to absorb the angular cliff, so the angular-moment closure over-corrects as M grows; the core, with two vacuum edges, converges on the reference instead.)

Flux comparison, explicit geometry. Figure 4 compares the pin-averaged fission maps $\Sigma_f \phi$ of the rectangular-node OpenPSN ($M=16$) against the OpenMC explicit-geometry (fuel circle + water annulus) runs on the same un-homogenised data, for the single UO_2 assembly (first row, 17×17 pins) and the 1/4 core (second row, 34×34 fuel pins). The OpenPSN node field is the pin power $\nu \Sigma_f \phi$; it is converted into the fission quantity by dividing each pin by the fission-weighted ν of its material (2.441 for UO_2 , 2.865–2.874 for the three MOX enrichments, and 2.434 for the fission-chamber inserts), so that both maps are the same physical quantity $\Sigma_f \phi$ pin by pin. Both maps are then normalised to unit sum, so the residual is purely in the shape. On the assembly the pin-by-pin relative error has a mean of -0.002% and an RMS of 0.070% (max 0.21%). On the 1/4 core, over all 1056 fuel pins, the error is a mean of -0.25% and an RMS of 0.72% (max 1.52%): the four assemblies show a coherent pattern at the $\pm 0.8\%$ level (UO_2 assemblies $+0.04 \dots +0.63\%$, MOX assemblies -0.84% in both) — the small between-assembly power redistribution that comes with the rectangular tiling — while the intra-assembly shape error stays below $\pm 1\%$ in every assembly; the two solvers agree within one percent on the pin-level fission distribution for the whole core.

4.5 Comparison with the published diffusion / SN / nodal landscape

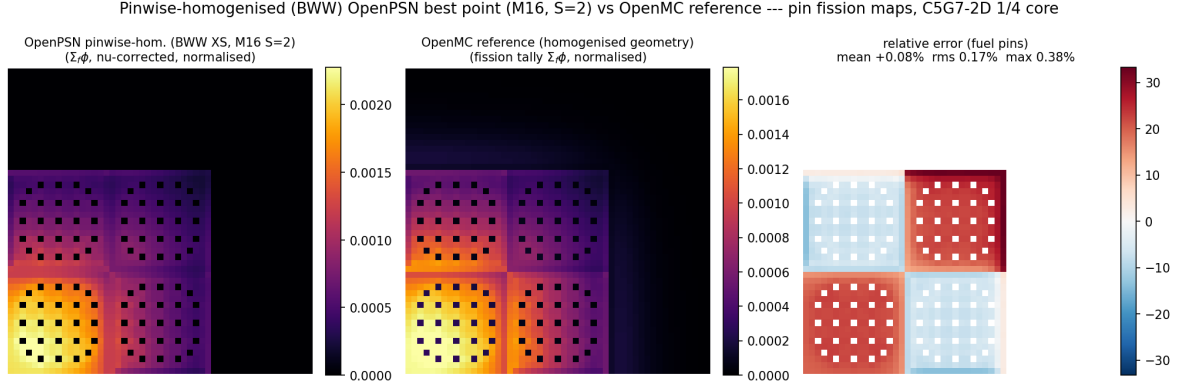


Figure 2: Pin-averaged fission maps $\Sigma_f \phi$ at the best pinwise-homogenised core point (M16, S=2) vs. the OpenMC homogenised-geometry reference on the same BWW XS, C5G7-2D 1/4 core (fuel region, 34×34 pins). Left: OpenPSN (ν -corrected); centre: OpenMC reference (7-group, collision); right: relative error (fuel pins only, both maps normalised to unit sum).

Rectangularisation of the C5G7 fuel rod: circle $\rightarrow$ equal-area square (pin pitch 1.26 cm and rod positions unchanged; only the centre rectangle of each rod carries the fuel material)

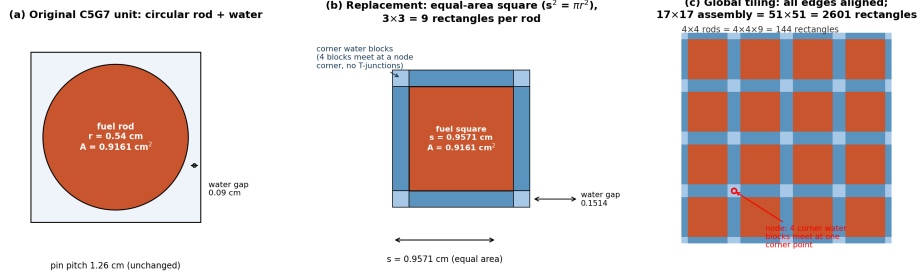


Figure 3: Rectangularisation of the C5G7 pin cell. (a) Original circular rod in its pitch cell (water gap 0.09 cm). (b) Equal-area fuel square ($s = 0.95713$ cm, $s^2 = \pi r^2$) with the pin cell split into 3×3 rectangles; the four corner water blocks of neighbouring pins meet at a single node corner, so the tiling has no T-junctions. (c) Global tiling: a 17×17 assembly is $51 \times 51 = 2601$ rectangles and the 1/4 core is $153 \times 153 = 23409$.

Table 3: Rectangular-node OpenPSN (un-homogenised material XS). Δ in pcm versus the explicitly noted reference. References: the directly computed OpenMC explicit-geometry (fuel circle + water annulus) runs on the same un-homogenised data, the high-fidelity reference 1.18646 of McGraw *et al.* [5], and nTRACER (SPHINCS).

	$M=8$	$M=12$	$M=16$
<i>Single UO₂ assembly, κ_{eff}</i>			
κ_{eff}	1.3337164	1.3340330	1.3343536
Δ vs OpenMC, explicit pin (1.3334947)	+22.2	+53.8	+85.9
Δ vs nTRACER (1.33367)	+4.8	+36.3	+68.9
<i>C5G7-2D 1/4 core, κ_{eff}</i>			
κ_{eff}	1.1853742	1.1861084	1.1866067
Δ vs OpenMC, explicit pin (1.1864955)	-112.1	-38.7	+11.1
Δ vs McGraw high-fidelity (1.18646)	-108.6	-35.2	+14.7
Δ vs nTRACER (1.18653)	-115.6	-42.2	+7.7

Rectangular-node PSN (M16) vs OpenMC --- pin fission maps $\Sigma_f \phi$ (both normalised)

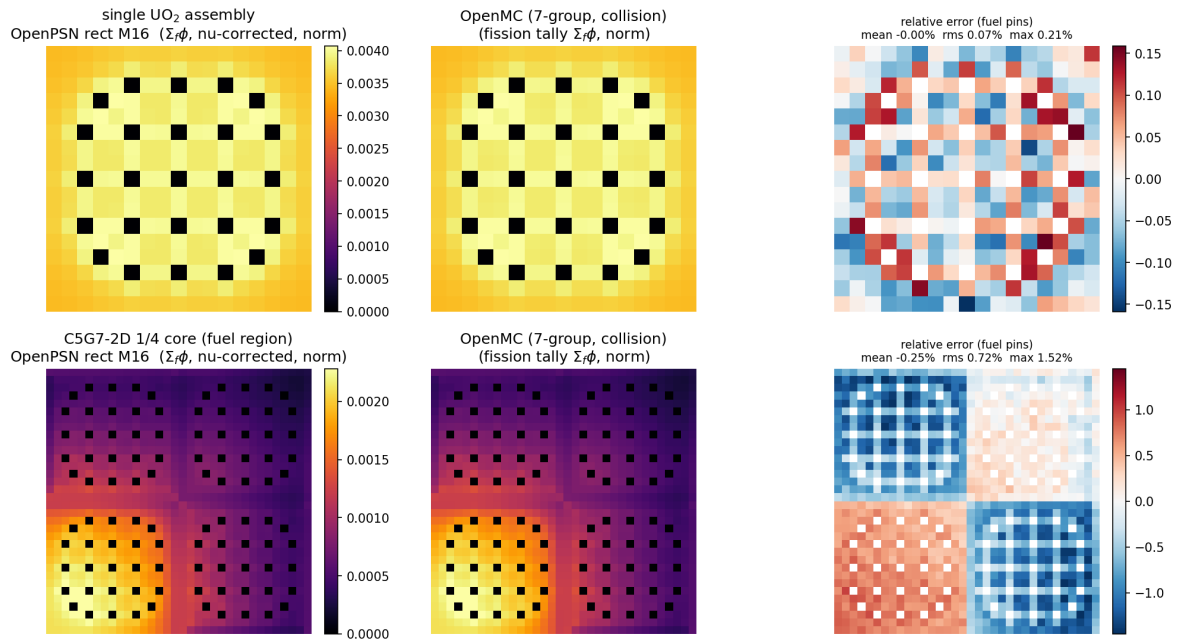


Figure 4: Pin-averaged fission maps $\Sigma_f \phi$, rectangular-node OpenPSN ($M=16$, un-homogenised material XS, ν -corrected) vs. OpenMC (explicit geometry, 7-group collision estimator). One row per problem — single UO₂ assembly (17×17 pins), then C5G7-2D 1/4 core, fuel region only (34×34 pins); columns are OpenPSN, OpenMC, and the pin-by-pin relative error (fuel pins only), both maps normalised to unit sum.

The original NEA report [2] tabulates 20 deterministic codes on C5G7-2D. Table 4 reproduces the diffusion- and nodal-family values we could extract from it, together with recent P1/SP3/SN/nodal literature results and our own OpenPSN result.

Table 4: C5G7-2D κ_{eff} by method class (2-D multi-group, homogenised pin model unless noted). NEA-2003 k values (its Table 17) plus recent P1/SP3 literature. $\Delta = (k - 1.18655) \times 10^5$ in pcm, except where stated (STELLA vs its PEACH reference 1.18716; OpenPSN vs 1.18646). The original NEA table prints the relative per-cent error; the pcm values here are recomputed from the printed k .

method class	code	κ_{eff}	Δ (pcm)
<i>Pure P1 diffusion</i>			
diffusion (FEM)	CRONOS2	1.18323	−332
P1 (pin-hom)	STELLA-P1 [9]	1.18426	−290 (vs PEACH)
<i>SN</i>			
SN-FEM	CRONOS2-SN	1.18338	−317
SN-FDM	DORT-GRS	1.18482	−173
SN-FDM	DORT-ORNL	1.18496	−159
SN-FDM	TWODANT	1.18668	+13
SN (diamond diff.)	PARTISN	1.18637	−18
SN (unstructured)	PERICLES	1.18658	+3
<i>High-order nodal P_n / surf. harmonics</i>			
P_n -nodal (FEM)	VARIANT-SE	1.18495	−160
P_n -nodal + int. transp.	VARIANT-ISE	1.18745	+90
surface harm. (G3/P2)	SUHAM-2D	1.18628	−27
<i>MOC</i>			
MOC	CHAPLET	1.18656	+1
MOC	MCCG3D	1.18657	+2
MOC	DeCART	1.18660	+5
MOC	APOLLO2	1.18618	−37
MOC	CRX	1.18813	+158
MOC (stoch. rays)	UNKGRO	1.18523	−132
<i>This work</i>			
PSN nodal (BWW pinwise-hom)	OpenPSN	1.1865870	+4 (vs 1.18655)
PSN nodal (rect. node, $M=16$)	OpenPSN	1.1866067	+5 (vs 1.18655)

Readings.

- **Pure P1 diffusion is systematically low** — CRONOS2 −332 pcm (−0.28%) vs 1.18655, STELLA-P1 −290 pcm vs its PEACH MOC reference 1.18716: no transport correction and no self shielding.
- **SN spans −317 (CRONOS2-SN) to +13 pcm (TWODANT); DORT −173/−159; the better-differenced / unstructured variants PARTISN (−18) and PERICLES (+3) close to $\pm 0.02\%$.**
- **High-order nodal P_n / surface harmonics** spans −160 (VARIANT-SE) to +90 pcm (VARIANT-ISE); SUHAM-2D −27; VARIANT-ISE also has the best pin power in the original table (MRE 0.11%).

- **MOC (explicit pin geometry) clusters within ± 37 pcm** (CHAPLET +1, MCCG3D +2, DeCART +5, APOLLO2 -37), with two outliers (CRX +158, UNKGRO -132); TWODANT is SN-FDM (+13); the high-fidelity Rattlesnake converged limit is 1.186446 (~ 1 pcm from the 1.186456 reference).
- **OpenPSN in this work.** The two C5G7 studies of Sections 4.3–4.4 bound the answer from both sides. With the self-shielding-aware BWW pinwise homogenisation, the best point of the $S \times M$ sweep ($M16_S2$) is -23.6 pcm vs. the OpenMC homogenised reference, i.e. $+4$ pcm vs. 1.18655; with the rectangular-node model that removes homogenisation entirely, $M=16$ lands at $+11.1$ pcm vs. the OpenMC explicit-geometry reference (inside its Monte Carlo standard error) and $+5$ pcm vs. 1.18655. In either case OpenPSN sits inside the MOC reference cluster (± 37 pcm, plus the better-differenced SN and unstructured codes), and the rectangular-node result is within the $M=16$ angular convergence of the nodal PSN method on this problem.

Two published outliers are worth noting for completeness: COHINT (P_2 interface currents, -1125 pcm) and HELIOS (collision probability, $+675$ pcm), both far outside the converged cluster and attributable to their respective current/transport approximations.

5 Conclusions

We built **OpenPSN**, a clean, multi-group, sparse, vectorised implementation of the Phase Space Nodal method, and validated it on two levels. On the source paper’s problems it reproduces all 248 published eigenvalue data points to a few pcm (full point-by-point comparison in Appendix A) while removing the dense-formulation memory blow-up (exact to 10^{-15} ; the 10404-node C5G7 case assembles in ~ 54 MB and converges in ~ 54 min on one core, where the dense form would need a ~ 504 GB matrix). We also extended the node operators to arbitrary per-node widths, giving the first PSN implementation with explicit, un-homogenised intra-node fuel/water structure.

On the C5G7 MOX benchmark the two studies separate the two error sources that a nodal code carries. In the *pinwise-homogenised* study, the $S \times M$ sweep against a directly computed OpenMC reference on the same homogenised data shows the discretisation converging cleanly: the assembly error reaches the Monte Carlo standard error (≈ 3.5 pcm) by $S=5$, $M \geq 8$, and the best core point (with the self-shielding-aware BWW recipe) is -23.6 pcm versus the OpenMC homogenised reference, i.e. $+4$ pcm versus 1.18655. That residual is the homogenisation model itself, and the best-point pin-power field agrees with the same-recipe OpenMC map to 0.17% RMS (max 0.38%): a smooth method-level bias, not a cancellation of local errors. In the *rectangular-node* study the homogenisation model is removed entirely: with un-homogenised 7-group material data the $1/4$ -core eigenvalue converges monotonically with M into the OpenMC explicit-geometry reference ($-112 \rightarrow -39 \rightarrow +11$ pcm for $M = 8 \rightarrow 16$), lands $+14.7$ pcm from the high-fidelity reference 1.18646, and its pin-averaged fission map $\Sigma_f \phi$ (ν -corrected, same quantity as the Monte Carlo tally) agrees with OpenMC to 0.07% RMS on the single assembly and 0.72% over all 1056 fuel pins of the core (max 1.52%; below $\pm 1\%$ within every assembly, largest in the MOX cells where the local gradients are steepest). Placed against the 20-code NEA landscape and the recent P1/SP3/SN/nodal literature, both OpenPSN results sit inside the high-fidelity MOC reference cluster: the nodal PSN method, once its homogenisation model is controlled (by a self-shielding-aware recipe or by explicit rectangular nodes), reproduces resolved-pin transport at a fraction of the mesh cost of discrete-ordinates or characteristics methods.

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A Full point-by-point reproduction of the published PSN results

Table 1 summarised the maxima. Tables 5–8 give the complete 240-cell comparison for the three printed eigenvalue grids (checkerboard weak/strong, BWR without Gd) plus the with-Gd valley, in the same “this work / paper (Δ) cell format used in the reproduction notes shipped in `snapshot/`. Row labels: S = spatial subdivision of the checkerboard grid, N = spatial grid of the BWR bundle. The remaining 8 points (restricted $M = 24$, $S = 1, 2, 4, 8$, weak/strong) agree with the printed Figs. 4–5 within the plot-reading uncertainty and are listed in `snapshot/data/report_data.json`.

Table 5: Fig. 3-A, weak absorption, generic ($I = 30$): Δ vs the printed values (reference $\kappa_{\text{eff}} = 1.12974$). S = spatial subdivision.

S	$M=2$	$M=4$	$M=8$	$M=12$	$M=16$	$M=24$
1×1	+136.2/136 (+0.2)	+389.0/388 (+1.0)	+703.7/703 (+0.7)	+793.9/793 (+0.9)	+835.2/835 (+0.2)	+873.5/873 (+0.5)
2×2	−275.4/ − 276 (+0.6)	−143.6/ − 144 (+0.4)	+154.9/154 (+0.9)	+217.2/217 (+0.2)	+244.4/244 (+0.4)	+268.2/268 (+0.2)
3×3	−374.4/ − 375 (+0.6)	−263.2/ − 264 (+0.8)	+19.1/18 (+1.1)	+79.3/79 (+0.3)	+104.4/104 (+0.4)	+124.1/123 (+1.1)
4×4	−412.0/ − 413 (+1.0)	−309.4/ − 310 (+0.6)	−34.6/ − 35 (+0.4)	+24.9/24 (+0.9)	+48.2/47 (+1.2)	+66.9/66 (+0.9)
5×5	−430.1/ − 431 (+0.9)	−332.1/ − 333 (+0.9)	−61.2/ − 62 (+0.8)	−2.7/ − 3 (+0.3)	+20.2/19 (+1.2)	+38.2/38 (+0.2)
6×6	−440.3/ − 441 (+0.7)	−344.8/ − 346 (+1.2)	−76.3/ − 77 (+0.7)	−18.5/ − 19 (+0.5)	+4.1/3 (+1.1)	+21.8/21 (+0.8)
7×7	−446.5/ − 447 (+0.5)	−352.7/ − 353 (+0.3)	−85.7/ − 86 (+0.3)	−28.4/ − 29 (+0.6)	−6.1/ − 7 (+0.9)	+11.4/11 (+0.4)
8×8	−450.5/ − 451 (+0.5)	−357.9/ − 359 (+1.1)	−92.0/ − 93 (+1.0)	−35.0/ − 36 (+1.0)	−12.9/ − 14 (+1.1)	+4.4/4 (+0.4)
9×9	−453.4/ − 454 (+0.6)	−361.5/ − 362 (+0.5)	−96.3/ − 97 (+0.7)	−39.6/ − 40 (+0.4)	−17.7/ − 18 (+0.3)	−0.5/ − 1 (+0.5)
10×10	−455.4/ − 456 (+0.6)	−364.1/ − 365 (+0.9)	−99.5/ − 100 (+0.5)	−43.0/ − 44 (+1.0)	−21.2/ − 22 (+0.8)	−4.1/ − 5 (+0.9)

Note: each cell is “this work / paper (Δ), Δ = this work − paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots (± 1 –2 pcm reading uncertainty).

Table 6: Fig. 3-B, strong absorption, generic ($I = 30$): Δ vs the printed values (reference $\kappa_{\text{eff}} = 0.51673$). S = spatial subdivision.

S	$M=2$	$M=4$	$M=8$	$M=12$	$M=16$	$M=24$
1×1	+591.0/591 (+0.0)	+1762.4/1763 (-0.6)	+3199.2/3199 (+0.2)	+3596.7/3597 (-0.3)	+3776.6/3777 (-0.4)	+3941.3/3941 (+0.3)
2×2	-1534.9/ - 1535 (+0.1)	-790.2/ - 790 (-0.2)	+807.9/808 (-0.1)	+1132.7/1133 (-0.3)	+1271.3/1271 (+0.3)	+1392.4/1393 (-0.6)
3×3	-2068.0/ - 2068 (+0.0)	-1425.6/ - 1425 (-0.6)	+113.2/113 (+0.2)	+431.2/431 (+0.2)	+563.2/563 (+0.2)	+666.4/666 (+0.4)
4×4	-2272.6/ - 2272 (-0.6)	-1676.0/ - 1676 (+0.0)	-170.0/ - 170 (+0.0)	+145.3/145 (+0.3)	+269.2/269 (+0.2)	+367.8/368 (-0.2)
5×5	-2371.6/ - 2372 (+0.4)	-1799.5/ - 1799 (-0.5)	-311.9/ - 312 (+0.1)	-1.4/ - 1 (-0.4)	+120.5/121 (-0.5)	+216.1/216 (+0.1)
6×6	-2427.0/ - 2427 (+0.0)	-1869.2/ - 1869 (-0.2)	-393.2/ - 393 (-0.2)	-86.1/ - 86 (-0.1)	+34.1/34 (+0.1)	+128.3/128 (+0.3)
7×7	-2461.0/ - 2461 (+0.0)	-1912.3/ - 1912 (-0.3)	-444.1/ - 444 (-0.1)	-139.3/ - 139 (-0.3)	-20.7/ - 21 (+0.3)	+72.6/72 (+0.6)
8×8	-2483.3/ - 2483 (-0.3)	-1940.8/ - 1941 (+0.2)	-478.0/ - 478 (+0.0)	-175.1/ - 175 (-0.1)	-57.6/ - 58 (+0.4)	+35.0/34 (+1.0)
9×9	-2498.8/ - 2499 (+0.2)	-1960.6/ - 1961 (+0.4)	-501.8/ - 502 (+0.2)	-200.2/ - 200 (-0.2)	-83.6/ - 84 (+0.4)	+8.4/7 (+1.4)
10×10	-2510.0/ - 2510 (-0.0)	-1974.9/ - 1975 (+0.1)	-519.1/ - 519 (-0.1)	-218.6/ - 219 (+0.4)	-102.6/ - 103 (+0.4)	-11.1/ - 12 (+0.9)

Note: each cell is “this work / paper (Δ), Δ = this work - paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots (± 1 -2 pcm reading uncertainty).

Table 7: Fig. 10, BWR 12-pin bundle without Gd (2-group): Δ vs the values read from the printed contour plot (reference $\kappa_{\text{eff}} = 1.18797$). N = spatial grid.

N	$M=4$	$M=8$	$M=12$	$M=16$	$M=20$	$M=24$
1	+126.4/123 (+3.4)	-74.8/ - 78 (+3.2)	-122.2/ - 125 (+2.8)	-140.3/ - 144 (+3.7)	-149.1/ - 152 (+2.9)	-154.2/ - 157 (+2.8)
2	+206.5/203 (+3.5)	+28.3/25 (+3.3)	-15.6/ - 19 (+3.4)	-31.9/ - 35 (+3.1)	-40.1/ - 43 (+2.9)	-44.9/ - 48 (+3.1)
3	+226.2/223 (+3.2)	+54.8/51 (+3.8)	+12.5/9 (+3.5)	-3.6/ - 7 (+3.4)	-11.6/ - 15 (+3.4)	-16.3/ - 20 (+3.7)
4	+233.7/230 (+3.7)	+65.2/62 (+3.2)	+23.8/20 (+3.8)	+7.8/5 (+2.8)	-0.1/ - 3 (+2.9)	-4.6/ - 8 (+3.4)
5	+237.4/234 (+3.4)	+70.4/67 (+3.4)	+29.4/26 (+3.4)	+13.6/10 (+3.6)	+5.8/2 (+3.8)	+1.3/ - 2 (+3.3)
6	+239.4/236 (+3.4)	+73.3/70 (+3.3)	+32.7/29 (+3.7)	+16.9/14 (+2.9)	+9.1/6 (+3.1)	+4.7/1 (+3.7)
7	+240.6/237 (+3.6)	+75.1/72 (+3.1)	+34.7/31 (+3.7)	+19.0/16 (+3.0)	+11.3/8 (+3.3)	+6.9/4 (+2.9)
8	+241.4/238 (+3.4)	+76.3/73 (+3.3)	+36.0/33 (+3.0)	+20.4/17 (+3.4)	+12.7/9 (+3.7)	+8.3/5 (+3.3)
9	+242.0/239 (+3.0)	+77.2/74 (+3.2)	+37.0/34 (+3.0)	+21.4/18 (+3.4)	+13.7/10 (+3.7)	+9.3/6 (+3.3)
10	+242.4/239 (+3.4)	+77.8/74 (+3.8)	+37.7/34 (+3.7)	+22.1/19 (+3.1)	+14.5/11 (+3.5)	+10.1/7 (+3.1)

Note: each cell is “this work / paper (Δ), Δ = this work - paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots (± 1 -2 pcm reading uncertainty).

Table 8: Fig. 12, BWR 12-pin bundle with Gd (2-group): this work only (the printed figure is a contour plot). N = spatial grid. Bold = best M in the row ($\min |\Delta|$); the valley shifts to lower M as N grows, exactly the Sec. 4.4 effect of the source paper. Best point overall: $N = 9$, $M = 20$, +1.1 pcm.

N	$M=4$	$M=8$	$M=12$	$M=16$	$M=20$	$M=24$
1	-605	+311	+565	+681	+746	+788
2	-998	-152	+67	+157	+203	+230
3	-1140	-253	-22	+73	+123	+152
4	-1207	-283	-45	+53	+103	+133
5	-1264	-318	-72	+28	+79	+110
6	-1310	-351	-98	+4	+57	+88
7	-1346	-380	-122	-18	+36	+67
8	-1375	-405	-144	-37	+17	+49
9	-1397	-427	-162	-54	+1	+33
10	-1415	-445	-178	-69	-13	+20

Note: each cell is “this work / paper (Δ), Δ = this work – paper, in pcm; paper values as printed in Ref. [1]; the Fig. 10/12 references are read from the printed contour plots (± 1 –2 pcm reading uncertainty).